

Computer Network Reference Models

Module 03

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hello everybody I am Professor yogesh Chaba from Guru jambheshwar University of Science and Technology harar harana today in this lecture we will study about computer network reference models this lecture on computer network reference models has been divided into three parts basic concepts OSI reference model and tcpip reference model let us start with computer networks computer network is a collection of autonomous computers interconnected by a single technology two computers are set to be interconnected if they are able to exchange information the connection between computers can be via a copper wire fiber optics microwaves infrared or communication satellite there is sometime confusion between a computer network and a distributed system the key distinction

is that in a distributed system a collection of independent computers appear to its users as a single coherent system often a layer of software on the top of operating system called middleware is responsible for implementing this model a well-known example of a distributed system is the worldwide web in which everything looks like a document in computer network different computers are connected through single Technology based upon coverage area computer network has been divided into four types first is pan that is personal area network second is Lan that is local area network third is man that is metropolitan area network and last is Van that is wide area network let us discuss each of them one

by one a pan is the interconnection of information technology devices within a range of around 10 M connecting a computer to wireless keyboard or a mouse or a printer or other computer comes under pan this kind of personal area network could also be interconnected with or without wires to the internet or or other networks most common pan technology available nowadays is Bluetooth Bluetooth uses short range radio waves over distances up to approximately 10 m next is local area network generally called lens they are privately owned networks within a single building or Campus of up to a few kilometers in size they are widely used to connect personal computers and workstations in company offices and

factories to share resources and exchange information. Lenses are restricted in size which means that the worst case transmission time is bounded and known in advance. Various topologies are possible for lenses. Figure shows two of them which are bus based and ring based. In a bus Network as shown in figure A on the left side, at any instant at most one machine is the master and is allowed to transmit. The orbitation mechanism may be 5 minutes, 1 second, centralized or distributed. i.e. 802.3 popularly called ethernet is a bus based broadcast network with decentralized control usually operating at 10 Mbps to 10 Gbps speed. A second type of broadcast system is the ring as shown in figure B on the right in a ring.

Network any computer which wants to communicate on ring captures a token and starts transmission. i.e. 802.5 the token ring is a ring based LAN operating at 4 and 16 Mbps speed. FDDI is another example of a ring Network. Next is MAN that is metropolitan area network, a network which can cover a city. The best known example of a MAN is cable television network available in many cities nowadays. WMAN is also coming up which is Wireless MAN technology. A MAN might look something like the system shown in this figure. We can see a MAN network in which both television signals and internet is being fed into the centralized headend for subsequent distribution to people's homes. Last type of network is WAN that

is wide area network. It spans over a large geographical area, maybe a state, a country or a continent. It contains a collection of hosts intended for running user programs. The hosts are connected by a communication subnet or just subnet. For short, the hosts are owned by the customers whereas the communication subnet is typically owned and operated by a telephone company or Internet service provider. The job of the subnet is to carry messages from host to host just as the telephone system carries words from speaker to listener. In most wide area networks, the subnet consists of two distinct components: transmission lines and switching elements. Transmission lines move bits between machines; they can be made of copper wire, optical fiber or even radio.

Links: switching elements are specialized computers that connect three or more transmission lines. In WAN as shown in figure, each host is connected to a LAN on which a router is present. The collection of communication lines and routers form the subnet, and collection of these lenses through a subnet spread across it. It is when after discussing types of network, let us move on to network architecture. We will discuss two important Network architecture: The OSI reference model and the TCP/IP reference model. Although the protocols associated with the OSI model are rarely used anymore, the model itself is actually quite journal and still valid. The features discussed at each layer are still very important. The protocols in TCP/IP reference model are

very widely used. The ISO's OSI model is based on a proposal developed by the International Standards Organization (ISO). It is the first step towards International standardization of the protocols used for communication purpose. OSI here stands for open system interconnection. This model is developed by ISO for systems which are open for interconnection. The OSI model

has seven layers the principles that were applied to arrive at the seven layers can be briefly summarized as follows a layer should be created where a different abstraction is needed each layer should perform a well defined function the function of each layer should be chosen with an eye to word defining internationally standardized protocols the layer boundaries should be chosen to minimize the information flow

across the interfaces the number of layers should be large enough that distinct functions need not be thrown together in the same layer keeping in view the reasons given OSI reference model is divided into seven layers starting from application presentation session transport network data link to physical layer all these seven layers are available at both transmitter and receiver side on transmitter side data from user is given to application layer it moves M layer by layer from application layer to physical layer and then to transmission media on receiving side data is received by physical layer from where it is passed on layer by layer to application layer user receives the data at application layer nomenclature of data units exchanged at each layer

are shown on figure at physical layer it is a bit at data link layer it is called frame at Network layer it is a packet at transport layer it is called tpdu that is transport protocol data unit at session layer it is sp duu then at presentation layer it is called PP du and at application layer it is called AP du another important aspect of this model is that there are two types of layers chained layers and end to endend layers lower three layers that is network data link physical are called chained layers protocols for chain layers are available at intermediate Network device in the network also these Protocols are between each machine and its immediate neighbors and not between the ultimate

source and destination machines which may be separated by many routers upper four layers application presentation session and transport are called end to endend layers these protocol protols are available only on end to endend computers now let us see function of each layer one by one lowermost layer that is physical layer is concerned with transmitting raw bits over a Communication channel the design issues have to do with making sure that when one side sends a one bit it is received by the other side as a one bit not as a zero bit job of physical layer is to decide about voltage level of bits timings for each bit modulation technique to be implemented and few others the design issues here largely deal with mechanical electrical bit

timing and the physical transmission medium which lies below the physical layer next layer is data link layer the main task of the data link layer is to transfer raw bits into a line that appears free of undetected transmission errors to the network layer it accomplishes this task by having the sender break up the input data into Data frames and transmit the frames sequentially if the service is reliable the receiver confirms correct reip of each frame by sending back an acknowledgement frame another issue that arises in the data link layer is how to keep a fast transmitter from drowning a slow receiver some traffic regulation mechanism is often need needed to let the transmitter know how much buffer

space the receiver has at the moment it's job of this layer to take care of flow control broadcast networks have an additional issue in the data link layer that is how to control access to the shared Channel a special sublayer of the data link layer known as Mac that is medium Access Control sublayer deals with this problem the network layer controls the operation of the subnet a key design issue is determining how packets are routed from source to destination if too many packets are present in the subnet at the same time they will get in one another way forming bottlenecks the control of such congestion also belong to the network layer addressing is also job of network layer more generally the quality of service provided is also a network layer

issue moving on to the transport layer the basic function of the transport layer is to accept data from above layer split it into two smaller units if need 17 minutes, 1 second be pass these to the network layer and ensure that the pieces all arrive correctly at the other end the transport layer also determines what type of service to provide to the session layer and ultimately to the users of the network the transport layer is a true end to end layer all the way from the source to the destination in other words a program on the source machine carries on a conversation with a similar program on the destination machine the session layer allows users on different machines to establish sessions between them sessions offer

various services including dialog control token management to prevent two parties from attempting the same critical operation at the same time and synchronization next layer is the presentation layer unlike lower layers which are mostly concerned with moving bits around the presentation layer is concerned with syntax and semantics of the information transmiss it also take care of encoding and decoding of data last layer that is application layer contains a variety of protocols that are commonly needed by users application layer protocols provides interface to the user one widely used application protocol is HTTP that is hyper text transfer protocol which is the basis for the worldwide web other application Protocols are for file transfer

electronic mail and network news so this was iso's OSI reference model now let us see the TCP IP reference model the TCP IP reference model is the network model used in the current Internet AR architecture it has its origin back in 1960s with the grandfather of the internet the arpanet this was a research Network sponsored by the Department of Defense in the United States the major design goals of tcpip reference model were ability to connect multiple networks together seamlessly ability for connections to remain intact as long as the source and destination machines were functioning to be built on flexible architecture the reference model was named after two of its main protocols TCP that is transmission control

protocol and IP that is Internet Protocol there are four layers in TCP IP reference model model as compared to 7 in OSI reference model host to network layer performs the jobs of data link layer and physical layer of OSI reference model internet layer is same as Network layer in OSI reference model for session and transport layer in OSI reference model there is only one layer

that is named as transport layer in tcpip reference model similarly for application and presentation there is application layer different protocols and networks in the initial version of TCP IP models are as shown in figure at Host to network layer there is arpanet setet Packet radio and Lan at Network layer there is IP

protocol that is Internet Protocol its job is to take care of routing addressing and congestion control at transport layer there is TCP and UDP TCP is reliable connection oriented and UDP is unreliable connectionless service their job is endtoend connectivity and flow control between and 22 minutes, 1 second users on top of Transport layer is the application layer it contains all the higher level protocols the early ones included virtual terminal tet file transfer protocol FTP and electronic mail protocol SMTP many other protocols have been added to these over the years like like the domain name system DNS for mapping host names onto their Network addresses mntp the protocol for moving uset news articles around and HTTP the protocol for fetching pages on the

worldwide web and many others The OSI and tcpip reference models have much in common both are based on the concept of a stack of independent protocols also the functionality of layers is roughly similar for example in both models the layer provide an endtoend Network independent Transport service to processes wishing to communicate again in both models the layers about trans Port are application oriented users of the transport service despite these fundamental similarities the two models also have many differences The OSI model completes the job using seven layers whereas tcpip model has only four layers the tcpip model did not originally clearly distinguish between service interface and protocol whereas OSI model does

so another difference is in the area of connectionless versus connection oriented communication The OSI model supports both connectionless and connection oriented communication in the network layer but only connection oriented communication in the transport layer but the TCP IP model has only connectionless mode in the network layer but supports both modes in the transport layer giving the user a choice so dear friends today in this lecture we have covered introductory concepts of computer networks followed by two very important reference models OSI and tcpip in last we have also seen comparison of these two models thank you